Use of Artificial Intelligence in the MyCycle Project

1. Introduction

Artificial intelligence (AI), mainly ChatGPT, was used as a support tool during the development of my MyCycle Android application. MyCycle is a menstrual cycle tracking application that allows users to record period dates, symptoms, moods and notes. I used AI to understand technical concepts, plan development tasks, solve programming problems and prepare project documentation. AI assisted me throughout the process, but I remained responsible for the implementation and testing of the application.

2. Planning and Understanding Requirements

I used ChatGPT to help me understand the assignment requirements and divide the project into manageable tasks. It helped me identify important features required for the final application, including connecting the Android application to a backend API, implementing Azure Blob Storage, using a NoSQL database, testing the application on a physical phone and preparing it for publication. This assistance helped me organise my development process and prioritise important requirements.

3. Programming and API Integration

During development, I used AI to explain Kotlin, Jetpack Compose and Retrofit. It helped me understand how the Android application communicates with the ASP.NET Core Web API. For example, I received assistance with creating request and response models and understanding how POST requests save information while GET requests retrieve information. AI also helped me structure Retrofit functions for saving and retrieving cycle records from the backend.

4. Azure Cloud Services

I used ChatGPT to understand how Azure Blob Storage works and how it can be used to store uploaded images. It explained concepts such as storage accounts, connection strings, containers and upload endpoints.

AI also helped me understand Azure Cosmos DB as a NoSQL database. I received guidance on configuring the database, creating the CycleRecords container, using the userId partition key and implementing operations to save and retrieve cycle records. I tested these operations through Swagger to confirm that the backend could store and return information.

5. Debugging and Testing

Whenever I experienced programming errors, I used ChatGPT to help identify possible causes and suggest solutions. This included errors involving ambiguous class references, API configuration, Swagger and Android integration. I shared relevant code and error messages to receive explanations.

However, I did not assume that every suggested solution was correct. I applied changes in my development environment, built the project and checked whether the errors were resolved. I also used Swagger to test backend functionality.

6. Responsible Use of AI and My Contribution

I remained responsible for developing the MyCycle application. I configured the projects and Azure services, reviewed and adjusted code, connected the application to the backend and tested its functionality.

I understand that AI-generated code can contain errors or may not match an existing project's structure. Therefore, I reviewed and adapted suggestions before using them. I also understand the importance of protecting sensitive information, such as passwords, user records and Azure connection strings.

Since MyCycle handles personal cycle-tracking information, user privacy and data security are important. The application's cycle predictions are estimates and should not be treated as medical advice or contraception.

7. Conclusion and Declaration

Using AI helped me improve my understanding of software development and overcome technical challenges during the MyCycle project. It was particularly useful for planning, programming assistance, debugging, cloud integration and documentation.

AI did not replace my responsibility as the developer. I remained responsible for reviewing, implementing and testing the suggested solutions.

I declare that I used AI as a supporting tool during this project and that I am responsible for the final work submitted. I will follow my institution's academic integrity requirements regarding the use of AI.